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Sacred forest biodiversity conservation: A meta-analysis

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Abstract

Sacred forests offer co-benefits of protecting both cultural traditions and forested areas. However, sacred forests' contribution to biodiversity conservation is often undervalued. Several reviews discuss biodiversity conservation in sacred forests, but large-scale studies that quantify the effect of sacred forests on biodiversity conservation are scarce. Many studies on the effectiveness of sacred forests in protecting biodiversity are limited to single-location censuses or lack comparisons against non-sacred forests that serve as controls. To quantify the impact of sacred forests on biodiversity conservation, we conducted a global meta-analysis that compares sacred forests with nearby non-sacred forests (i.e., control areas). Using 35 studies from 17 different countries, we found that sacred forests harbored similar levels of biodiversity as nearby forested areas. When comparing taxonomic groups, we found that the positive benefits to biodiversity in sacred forests compared to non-sacred forests were higher for plants compared to non-plant taxa. Our meta-analysis provides quantitative evidence that sacred forests can be effective areas of biodiversity conservation. Based on our results, we suggest that researchers interested in sacred forest biodiversity compare sacred forests with other land use types, collect standardized metadata from sacred forests and nearby areas that serve as comparisons, and extend monitoring to include more non-plant taxa. Sacred forests can preserve ecosystem function, provide social benefits, and play a role in fighting against climate change, and should not be overlooked.

KEYWORDS

biodiversity, conservation biology, meta-analysis, protected area, sacred forest

1 | INTRODUCTION

Forests throughout the world are becoming increasingly fragmented, with negative and significant impacts on biodiversity (Collinge, 2009; Haddad et al., 2015). Global habitat loss leads to smaller, more isolated fragments, with forest edges exposed to high-impact land uses, ultimately reducing ecosystem connectivity and decreasing populations (Lindenmayer and Fischer, 2006). Therefore, there is a need for creative and effective strategies to protect biodiversity and maintain ecosystem function in these fragmented landscapes. Sacred forests—forests with religious significance to local communities—can provide refuge for biodiversity in fragmented, human-modified

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landscapes (Dudley et al., 2010). Typically, sacred forests are situated around a point of worship, such as a shrine or church (Dudley et al., 2010), and are usually managed by local people (e.g., Indigenous groups or local communities) who act as stewards of the surrounding forest (Wild et al., 2008). The biodiversity conservation conferred by sacred forests can thus complement biodiversity conservation efforts in formally recognized protected areas that are more often managed by governments or NGOs (Klepeis et al., 2016).

Recognizing the conservation value of small-sized protected areas (Volenec and Dobson 2020) and community-managed areas (Verschuuren, 2007; Wild et al., 2008) has been a challenge. The International Union for the Conservation of Nature (IUCN) recognizes sacred forests as Indigenous and Community Conserved Areas (ICCAs) (IUCN, 2009). However, many ICCAs are not officially recognized as protected areas by national governments, which means that forest management by Indigenous peoples and local communities often goes unrecognized. For instance, Indigenous peoples or local communities only govern 0.75% of official protected areas as reported by the World Database of Protected Areas (WDPA); in comparison, one third of the global land surface is occupied, managed or used by Indigenous communities, and about 40% of terrestrial protected areas overlap with Indigenous lands (Garnett et al., 2018). Acknowledging that community-managed areas are important refuges for biodiversity, highlights bottom-up conservation efforts globally and their contributions to biodiversity conservation.

Sacred forests can offer significant benefits. First, they can protect biodiversity as refuges for rare and endemic species—in certain contexts, small patches of forest can maintain up to 85% of native species richness (Rösch et al., 2015). Second, sacred forests preserve forested habitats that can facilitate forest regeneration outside the sacred area (Mgumia and Oba, 2003; Bhagwat et al., 2005b). Third, given the importance of sacred forests for local communities, they can successfully maintain their biodiversity for a long time, as the cultural value of the forest can outlast one human generation (Manna and Roy, 2021). Fourth, sacred forests protect important ecosystem services and functions, such as protecting medicinal plants (Mgumia and Oba, 2003; Bhagwat et al., 2005a; Ma et al., 2022), ensuring landscape connectivity as wildlife corridors or buffer zones for protected areas (Ishii et al., 2010), facilitating seed dispersal and pollination (Rajasri et al., 2017), providing erosion control and protecting water resources (Ma et al., 2022), and mitigating some negative effects of climate change by providing temperature control (Rawat, 2014) and carbon storage (Waikhom et al., 2018; Devi et al., 2021). Finally, despite their small size, sacred forests can play a crucial role in a land-use matrix: overlooking this role would be a missed opportunity (Bhagwat et al., 2005b; Dudley et al., 2010; Marks et al., 2020).

Many theories of conservation (e.g., fortress conservation) frame humans as detrimental to nature (Montgomery et al., 2020). However, human management and interactions with nature do not inherently produce negative results (Kimmerer, 2013; Zeng, 2018). Humans have a long history of interacting with forests without negatively impacting their biodiversity, such as in old-growth forests in New England (Mclachlan et al., 2000), in boreal forests in Sweden (Josefsson et al., 2009), and in tropical forests of Central Africa (Morin-Rivat et al., 2017) and Amazonia (Levis et al., 2017). Moreover, both forests and people have diverse histories, shifting over time. Sacred forests have had variable trajectories: some are remnant patches of older, formerly continuous forest (Ramanujam and Kadamban, 2001; Scull et al., 2017), while others appear to be patches of regenerated forest (Bhagwat et al., 2014). Similarly, groups of people are highly diverse and dynamic, with different religions and practices that determine how management is carried out (Li, 1996; Dove et al., 2011). Sacred forests are a prime example of the dynamic interaction between humans and nature.

Sacred forests are unequivocally important for the protection of cultural values and natural assets within and across geographic areas and among different cultures. For example, sacred forests can provide important resources: firewood, medicinal or ceremonial plants, and nontimber forest products (e.g., fruits and seeds). Additionally, sacred forests act as a setting for spiritual, religious, and cultural activities such as prayer, ceremony, and connecting with ancestors (Lynch et al., 2018; Ma et al., 2022). However, it is not clear what the quantitative impact a sacred status and associated forest management has on biodiversity, compared to forests without a sacred status. Sacred forest management could confer biodiversity benefits: stewards of sacred forests may not only maintain high habitat quality and limit chronic and acute forest disturbance (as remnant forests; Ramanujam and Kadamban, 2001; Scull et al., 2017), but also facilitate passive restoration (as regenerating forests; Bhagwat et al., 2014). Alternatively, sacredness may not translate to biodiversity protection: the management or use of sacred forests may create disturbance effects similar to (or greater than) other non-sacred forests nearby (Ormsby, 2011; Fraser et al., 2016).

The lack of comparative data from sacred and non-sacred forests from a variety of sites hinders our understanding of the benefits that sacred forests may provide

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for protecting biodiversity. Without a standardized comparison between sacred and non-sacred forests, we are unable to quantify the impact that sacred forests have on biodiversity (Mgumia and Oba, 2003). This knowledge gap may not be a result of a lack of interest or effort in the scientific research: local communities may decide against external scientific mapping or assessment of their sacred forests, rendering them 'invisible'. This invisibility may be an advantage for some communities and sacred forests because it is less likely they will become targets of extraction and because it will insulate them from the consequences of becoming formally recognized by top-down governance initiatives (see: Alusiola et al., 2021). Previous reviews have summarized the importance of sacred forests for conservation (Dudley et al., 2010; Zannini et al., 2021). Extending from such findings and using comparative data across different regions across the globe, we provide here a quantitative meta-analysis to determine the direction, magnitude, and significance of the impacts that sacred forests have on biodiversity conservation (Vetter et al., 2013). In our study, we compared species diversity from paired sacred and non-sacred forests across taxonomic groups, political boundaries, and protection status to evaluate the quantitative impacts of sacred forests on biodiversity conservation. We expected that if sacred forests were not effectively conserving biodiversity, sacred forests would have lower levels of biodiversity than both protected and unprotected non-sacred forest controls. Alternatively, if sacred forests effectively conserve biodiversity, we expected they would show similar or higher levels of biodiversity than controls.

2 | METHODS

2.1 | Study search

We collected data from published articles investigating biodiversity in sacred forests. We looked for articles on Web of Science that, as of April 2020, matched our search query: (sacred forest OR sacred grove OR church forest OR church garden) AND (biodiversity OR species richness OR diversity). The search scanned for articles with the search terms in the title, abstract, and keywords of the article. We found 631 articles that matched our criteria. Then, we examined the main text of each article, only selecting studies that measured species diversity using any appropriate metric for one or more taxonomic groups within sacred forests as and within appropriate control forests located nearby controls were any non-sacred forest area of a similar ecosystem type where species diversity was surveyed using similar methods to those in nearby sacred forests. We excluded studies that compared sacred forests with non-forested land. We ensured that data from controls and sacred forests came from the same article. Sacred forest is a term typically afforded to mostly natural wooded sacred sites, but we extended this term to also include some grasslands considered sacred by local communities because our search revealed studies from non-forested landscapes as well (e.g., Brandt et al., 2013; Deák et al., 2020). Studies were only included if they contained multiple samples from sacred forests and multiple samples from corresponding controls because obtaining a standard deviation from multiple samples was necessary to calculate effect sizes. We restricted our study to articles published in English and in peer-reviewed journals. We supplemented our search by adding additional records from a set of 27 studies from a recent systematic review of sacred natural sites by Zannini et al. (2021). This workflow resulted in 35 articles that were then used for subsequent analysis (Table S1). We acknowledge that studies published in other languages and gray literature about the topic do not appear in our search. However, we believe that our search method was wide enough to capture a representative sample of studies that have assessed some measure of biodiversity in sacred forests and in forest areas nearby (i.e., controls).

2.2 | Data collection

For each of the 35 articles, we extracted the mean, standard deviation, and sample size for any diversity metric (defined below) of sacred forests and corresponding controls. If a study compared more than one taxonomic group or had a paired design where a specific sacred forest was compared to a specific control, these comparisons resulted in multiple data points (hereafter records) per article. Following these criteria, we extracted a total of 63 records from the articles. We included a record regardless of the metric used to characterize diversity as long as both the sacred forests and control sites were measured with the same metric. 50 records (79%) used counts of the number of species (recorded as observed richness or species density); five records (8%) used Shannon diversity; Chao-1, Margalef estimated richness, and density of Operational Taxonomic Units were used in two records each; and rarefied richness and an acoustic metric (Acoustic Complexity Index) in one study each. For each study, we also recorded the study region, country, the focal taxa, and whether the control area was considered an official protected area or not, if this information was provided. One study did not report a standard deviation for either sacred forests or controls. In this case, we followed Bracken (1992) and imputed the standard

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TABLE 1 Summary of the literature included in the meta-analysis. A) The number of records per country where an aspect of biodiversity has been compared between a sacred forest and a nearby forested area (i.e., control) and B) records per taxonomic group in the dataset.

A. Country	Record count
Africa	
Ghana	8
Ethiopia	2
Nigeria	2
South Africa	2
Guinea Bissau	2
Cameroon	1
Mozambique	1
Liberia	3
America	
Mexico	1
Asia	
India	18
Thailand	3
China	2
Tibet	2
Tanzania	1
Europe	
Hungary	5
Italy	1
Greece	9
B. Taxonomic group	**Record count**
Plants	41
Invertebrates	10
Vertebrates	7
Fungi	2
Lichens	2
Bacteria	1

deviation using the coefficient of variation of other studies in the dataset with complete cases. For imputation, we used the metagear::impute_SD() function in the R package metagear (Lajeunesse, 2016).

2.3 | Statistical analysis

To test whether sacred forests contain equal, greater, or lesser diversity than nearby forested areas, we calculated the effect size as the standardized difference between sacred forests and their corresponding control. We used the Hedges-G metric, bias-corrected for heteroskedasticity in the population variance of the two groups (Hedges, 1981), by employing the 'escalc()' function in the 'metafor' R package (Viechtbauer, 2010). We also tested whether differences in diversity between sacred forests and control areas were associated with the protection status afforded to the control forest by including the protection status of the control forest as a fixed effect. Finally, we tested whether differences in diversity between sacred forests and control areas varied among taxa by broadly grouping records into plant versus non-plant records. In all cases, we used multivariate linear mixed-effects models, using study ID as a random intercept to account for pseudo-replication from multiple pairs of the same taxon in a single study and pseudo-replication from multiple taxa measured in a single study. We implemented this model using the 'metafor::rma.mv()' function (Viechtbauer, 2010). We ran all data screening, manipulation, analysis, and visualization using the *ggplot* and *dplyr* (Wickham et al., 2015), *metagear* (Lajeunesse 2016), and *metafor* (Viechtbauer, 2010) packages in R v4.0.4 (R Core Team 2021).

3 | RESULTS

3.1 | Dataset summary

Our final dataset encompassed 63 records from 35 studies conducted in 17 countries and across several different taxonomic groups (Table 1). The majority of records came from tropical or subtropical regions.

3.2 | Meta-analysis of species richness

Across all studies, the effect sizes of species diversity were similar between sacred forests and control sites (standardized mean difference [SMD] = 0.18, −0.27 to 0.63 95% confidence interval) (Figure 1). However, we observed large variation in the differences between sacred forests and control forests across studies, ranging from −7.12 to 16.27 SMD (QE = 394.0, $p < .0001$, Figure 1). We found no evidence of publication bias using a rank correlation test of asymmetry in the funnel plot (Kendall's tau = 0.128, $p = .14$, Figure S1). Moreover, we did not find a statistically significant effect of the protection status of control sites on the difference in diversity between sacred forests and controls (QM = 0.23, d.f. = 2, $p = .89$). Both protected and unprotected controls had effect sizes that overlapped with zero (Figure 2a).

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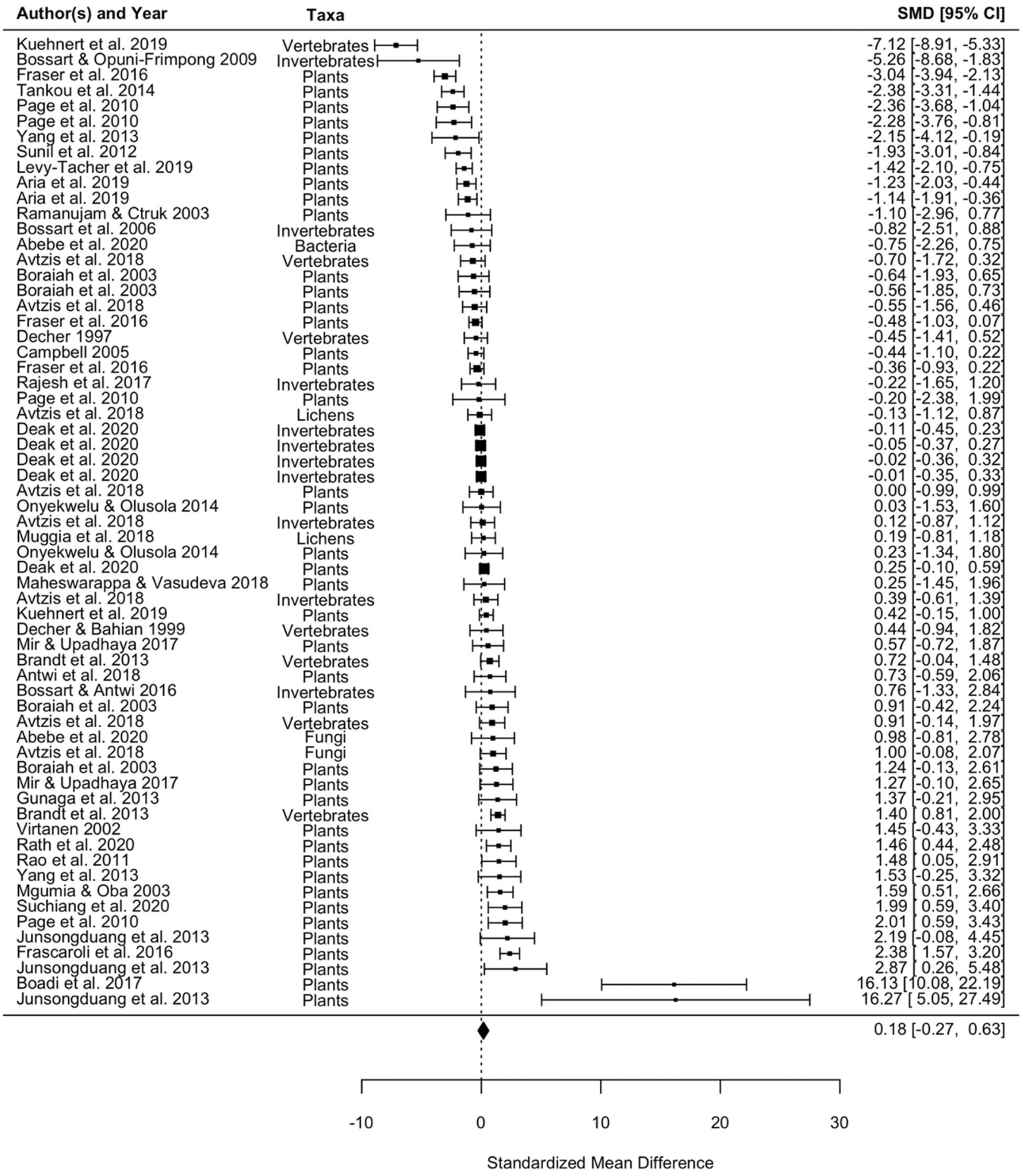


FIGURE 1 Results of meta-analysis estimating the effect size and 95% confidence interval (CI) between species diversity in sacred forests versus control sites across 63 records and 35 studies. Effect sizes were calculated as a standardized mean difference (SMD) between sacred forests and control sites. Negative effect sizes indicate lower species diversity in sacred forests compared to control sites. The size of each point is proportional to the precision of the effect size estimate (inverse standard error), with larger points indicating higher precision.

The estimated effect of sacred forests compared to controls was different for plants versus non-plant taxa (QM = 11.41, d.f. = 2, $p = .0033$, Figure 2b). For plants, diversity tended to be higher in sacred forests compared to controls (SMD = 0.41, −0.06 to 0.88 95% CI), whereas, for non-plants, diversity tended to be lower in sacred

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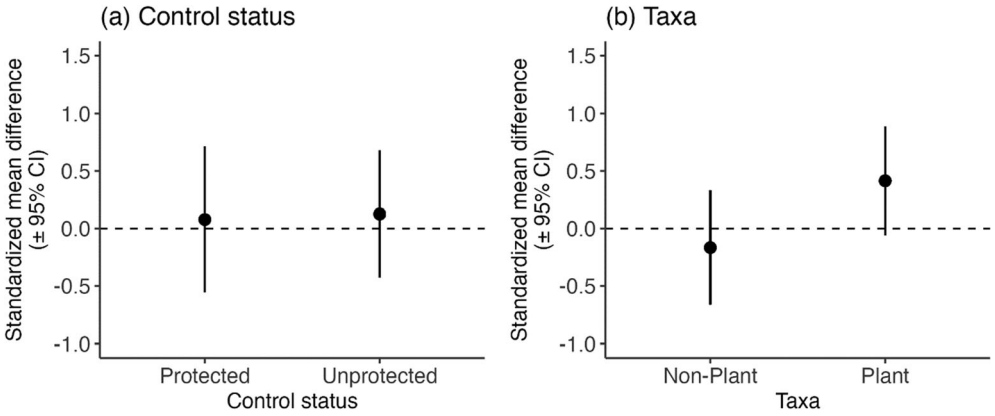


FIGURE 2 Effect sizes (standardized mean difference) and 95% confidence interval (CI) estimated separately for (a) whether control sites were Protected or Unprotected and (b) plant and non-plant taxa. The dashed line shows a standardized mean difference = 0, indicating no difference in species diversity between sacred forests and controls.

forests (SMD = −0.17, −0.66 to 0.33 95% CI). Nevertheless, the 95% confidence intervals in both cases overlapped zero and there was a high amount of uncertainty in these estimates (Figure 2b).

4 | DISCUSSION

4.1 | Sacred forests are as diverse as non-sacred forests

We found that sacred forests harbor similar quantitative levels of biodiversity compared to nearby non-sacred forests. This means that sacred forests critically contribute to biodiversity conservation, just as other studies have highlighted the importance of shrines, sacred natural sites, and the role of Indigenous and local communities in conservation (Reynolds et al., 2016; Frascaroli et al., 2019; Zannini et al., 2021; Dawson et al. 2021). While recent studies have reviewed the literature and stressed the importance of sacred forests in biodiversity conservation (e.g., Zannini et al. 2021), our study provides a quantitative and standardized test to adequately compare sacred forests with nearby forested areas. Even though we were not able to compare compositional differences between species with our dataset, we note that sacred forests are likely to be managed for plants of value (see below).

Unfortunately, we were unable to assess the impact of patch size or isolation had on sacred forest versus non-sacred forest biodiversity because information on forest size was either not reported or locations and boundaries of sacred forests were not published. Indeed, larger sacred forests are expected to have fewer edge effects and connectivity with other forests, sacred or otherwise, would ameliorate extinction risk (Laurance et al., 2009; Gibson et al., 2011). Patch shape, edge-to-interior ratio, and proportion of vegetation cover across the landscape, among other factors, would also help further explain the variations in biodiversity between sacred forests and nearby forested sites (Murcia, 1995; Harrison, 1997; Fischer & Lindenmayer, 2002; Radford et al., 2005). However, these data were not provided in the literature we reviewed. We acknowledge that there can be significant costs to local communities if the locations, boundaries, configuration, and extent of sacred forests are openly available: historical land-tenure conflicts can be exacerbated, and control over who is allowed access to the sacred forests can become difficult (Li, 1993). With these potential negative externalities in mind, researchers should carefully consider the potential impact before making efforts to collect new and additional data about sacred forests. One approach might be where, similar to that adopted for studies of rare species, key locations and other details are not reported publicly when data is collected, and studies are published.

4.2 | No impact of the protected area status of control forests

We found that sacred forests maintained similar levels of biodiversity when compared to both protected control areas and unprotected control areas. These results suggest that sacred forests have a similar effectiveness at maintaining biodiversity as comparable protected areas as well as comparable unprotected areas. However, the effectiveness of protected areas at conserving biodiversity on a global scale is variable and dependent on the local and regional context, with up to 40% of protected areas showing some form of deficiency (Naughton-Treves et al., 2005; Leverington et al., 2010). Sacred forests can reinforce larger scale protected area networks and fill in gaps that those protected areas may not cover (Dudley et al., 2009; Frascaroli et al., 2019; Zannini et al., 2021). It remains unknown whether the protected controls in our meta-analysis studies were effective at conserving biodiversity compared to unprotected controls. However, the

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difference in effect sizes between sacred forests and protected non-sacred control areas was positive and slightly higher than the difference in effect sizes between sacred forests and unprotected non-sacred control areas. Such differences may indicate that forests with protected areas close to them are experiencing a positive effect from higher-quality surrounding landscape matrix (Volenec and Dobson, 2020). Characterizing distance to nearest protected area or contiguous forest and quality of the landscape matrix would help quantify spillover effects of the sacred forests' surroundings. Additionally, the unprotected, non-sacred forests in our meta-analysis may simply be forests where pressure by human activity is very low, and we would expect results to be different if the unprotected areas had higher human pressures or were of different land-use types. Collecting descriptive metadata about the control sites would help tease apart which of these patterns may be occurring.

4.3 | Sacred forests have higher plant diversity than non-sacred forests

We found that sacred forests have higher mean plant diversity than non-sacred forests, but non-plant taxa have lower mean diversity in sacred forests than in non-sacred forest sites. One possible explanation for higher plant diversity in sacred forests could be that local people augment plant diversity by actively planting species that may be absent in nearby forested sites (Barrow, 2012). Medicinal plants and plants of use (e.g., plants harvested for food or ceremonial use; Belcher and Vantomme, 2003) are often found in sacred forests even when those species are missing from nearby ecosystems—thus increasing the overall plant diversity at these sites (Khan et al., 2008; Nganso et al., 2012, Verschuuren et al., 2012). Despite the high variability in our results, lower non-plant diversity in sacred forests may indicate delayed species extinctions (i.e., extinction debt) due to habitat disturbance (Kuussaari et al., 2009), or that fauna avoid sacred forests due to the prevalence of human use. Extinction debts have been found in many other fragmented forests (Alcocer-Rodriguez et al., 2021), and animals might avoid sacred forests due to their proximity to roads or human-induced noise (Laurance et al., 2009; Burivalova et al., 2021). Moreover, subsistence hunting may negatively impact animal diversity in some sacred forests (Lynch et al. 2018). Nevertheless, recent studies indicate that ecosystem services such as pollination and dispersal (Ricketts, 2004; Ismail et al., 2017) are retained in sacred forests (Bossart and Antwi, 2016, Woods et al., 2020). These results highlight the importance of non-plant taxa in sacred forests and emphasize the importance of monitoring non-plant biodiversity that maintains the ecological function of forests. A better understanding of the cultural uses of sacred forests (Zannini et al., 2021; Ma et al. 2022) and what activities are allowed and prohibited would help to better understand the variable impacts that sacred forests have on different taxa.

4.4 | Regional bias indicates a knowledge gap

We did not have statistical power to test how the geographic location of studies affected the effectiveness of sacred forests for biodiversity conservation. The highest concentration of studies per country in our dataset was in India, followed by Ghana. In a review of biodiversity studies from Africa and Asia (Dudley et al., 2010), India was noted as having the best-surveyed sacred sites, between 100,000 and 150,000 (Dudley et al., 2010). However, based on our literature review, many have not been compared to non-sacred forests nearby. This regional bias indicates that our findings are more generalizable to Asian, African, and European forests, and less applicable to forests in the Americas. For instance, we found only one study contrasting diversity in a sacred forest and a non-sacred forest from the Americas. Moreover, we observed that researchers in the Americas less commonly use the terms "sacred forests", "sacred groves" or "sacred natural sites" to describe vegetated areas used to express a relationship between local people and the divine, or areas around points of worship. One potential reason for this is that religious conversion has contributed to changes in the cultural ecological landscape in the Americas (Steinberg, 2002). Literature about community forests in the Americas often focuses on traditional agroforests, including the one case study we found in our meta-analysis from Mexico (Levy-Tacher et al., 2019). The low number of studies from the Americas highlights a limitation in the language used by ecologists and conservation scientists to denominate these managed areas and recognize their importance, across different regions and cultures (Dove et al., 2011; Dudley et al., 2010; Zannini et al., 2021). There are studies from Central and South America recognizing the value of sacred landscapes (Castro and Aldunate, 2003; Garcia-Frapolli et al., 2007). Sacred forests equivalents are also mentioned in the debate between land-sharing versus land-sparing when considering agricultural development and the effect on Indigenous communities and landscape management using traditional practices (Oliveira and Hecht, 2016). However, these studies are much broader in their approach and do not provide biodiversity measurements at specific sites in sacred forests equivalents, nor do they make comparisons with non-sacred forest areas. In North America, cemeteries and

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churchyards have been noted as havens of biodiversity (Löki et al., 2019), although they are not referred to as "sacred groves" or "sacred forests." These discrepancies highlight the importance of language used when discussing the links and connections between religion and ecology and addressing regional biases that these language differences confer.

4.5 | Evaluating the impact of conversion can provide information for decision-making

As scientists who study diversity and consider the benefits of sacred forests, we must ask not only *what* the benefits of sacred forests are, but also *who* these benefits serve (Montgomery et al., 2020). Sacred forests are areas that are protected, but because they are often not recognized as official protected areas—they are not included in most protected area maps or are devalued because of their small size. They may be also targeted for conversion into other types of land use because they lack an official protection status. Conversion may be driven by many different factors. Land may be converted without the consent of local communities due to existing land tenure conflicts, lack of local tenure recognition, and unequal power dynamics between industry, government, or outside stakeholders, and the local communities (Li, 1999; Ormsby, 2011). Moreover, sacred forests may be particular targets for imposed conversion into agroforestry, plantations, logging concessions, mines, or urban development (Daye and Healey, 2015) because they are often accessible (e.g., close to cities, towns, or villages). There also may be conflicts with land tenure or land rights (i.e., are able to be claimed/used by the community), or they are viewed as small, degraded fragments (Kandari et al., 2014). Local communities themselves may also consider sacred forests as sites of potential conversion or use for various reasons, including cultural shifts and to support livelihoods (Ormsby, 2011) or in response to changes in local polices on land use (Brandt et al., 2015).

Broadly speaking, land-use change can reduce biodiversity globally and across taxa (Gibson et al., 2011), and many production landscapes can have negative impacts on biodiversity and ecosystem function (Daye and Healey, 2015; Davis et al., 2020). If land conversion is being considered for livelihood reasons by local communities, information about returns on investment may help clarify assumptions about the expected financial gains and could help avoid regret after conversion (Ormsby, 2011). For example, it would be crucial to estimate how long it will take for lands to become income-generating, and how long the land will remain income-generating after conversion (Sheil and Wunder, 2002). Our results highlight a potential knowledge gap concerning how conversion decisions are made, and the impacts on the ecosystem and local communities after a land conversion occurs.

5 | FUTURE RESEARCH DIRECTIONS

We directly quantified the differences in biodiversity between sacred forests and non-sacred forests. If more metadata is provided in studies that describe sacred forests and nearby sites, future studies could compare how different land-use types and landscape matrices modulate the strength by which sacred forests at preserving biodiversity (Volenec and Dobson, 2020). For example, comparing sacred forests, protected area controls, and unprotected area controls within the same ecosystem would resolve the degree to which a designation of sacred forest and protected area is associated with differences in biodiversity compared to unprotected areas. Additionally, to understand how biodiversity and ecosystem services could be impacted by forest conversion, future studies could directly compare biodiversity between sacred forests and various production landscapes (e.g., agroforestry, Sharma and Vetaas, 2015).

Non-plant taxa were underrepresented in our dataset. Monitoring wildlife is a key component of conservation (Nichols and Williams, 2006). Information on what animal species use sacred forests could be useful for additional applications of interest to local managers, such as ecotourism in sacred forests (Takahashi et al., 2012; Tiimub et al., 2020) and information on other organisms (e.g., insects) can provide valuable information about ecosystem services such as pollination (Sitotaw et al., 2022). Information on non-plant taxa could inform management efforts and could help identify where support can be most effectively employed, in line with the goals of the local community.

A limitation of our study is that we did not undertake place-based research, and instead relied on existing, published information about sacred forest biodiversity. Moreover, new research that addresses if and how to include sacred forests into conservation strategies should meaningfully engage with the communities that manage them, as sociocultural and political context may be different from place to place (Allendorf et al. 2014). Designing studies with the input of local experts and communities is critically important (Montgomery et al., 2020). Integrating the local cultural and political context is needed to make decisions about the appropriateness of further data collection—for example, what data can and should be collected and how and in what ways further study can

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help strengthen relationships and knowledge exchange between Indigenous and local communities, and ecologists and conservation practitioners. Finally, though further data collection could serve the scientific community to address biases that we were unable to explore (e.g., patch size, distance from other protected areas, quality of the surrounding landscape matrix) and further contribute to understanding biodiversity conservation in sacred forests, we stress that not all sacred forests should be data points in a meta-analysis. Respecting traditional knowledge and the sovereignty of Indigenous and local communities (Latulippe and Klenk, 2020), and honoring decisions of refusal (Birch, 2018) are paramount.

AUTHOR CONTRIBUTIONS

Megan K. Sullivan: conceptualization, data curation, methodology, writing—original draft, review & editing; Luke Browne: data curation, methodology, visualization, writing—review & editing; Juan Carlos Penagos Zuluaga: data curation, writing—review & editing; Jasmine Liu: data curation, writing—review & editing; Akshay Surendra: data curation, methodology, writing—review & editing; Sergio Estrada-Villegas: data curation, methodology, writing—review & editing.

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DATA AVAILABILITY STATEMENT

The meta-analysis dataset from our study will be submitted to an online repository by the corresponding author (Megan K. Sullivan) upon acceptance of the manuscript.

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